



Measurement of Central Tendency

Your Complete Beginner-Friendly Reference Guide

Statistics for Data Science & Machine Learning

💡 What is Central Tendency?

Imagine you have a class of students and want to describe all their marks with JUST ONE NUMBER. That single representative number — the 'center' of your data — is called the Measure of Central Tendency.

Think of it like asking: 'What is the TYPICAL score in this class?'

There are multiple ways to find this center — each has its own strength and use-case. Choosing the wrong one gives you a misleading picture of your data!

Quick Overview — All 5 Measures

Measure	Best For	Weakness	Outliers?
Arithmetic Mean	General average, symmetric data	Pulled badly by outliers	✗ Sensitive
Geometric Mean	Growth rates, ratios, percentages	Cannot use with 0 or negative values	⚠ Somewhat
Harmonic Mean	Speeds, rates, ratios	Cannot use with 0 values	✓ Better
Median	Skewed data, data with outliers	Ignores magnitude of values	✓ Yes
Mode	Most frequent value, categories	May not exist or may be multiple	✓ Yes



Key Rule: $\text{Harmonic Mean} \leq \text{Geometric Mean} \leq \text{Arithmetic Mean}$

This is ALWAYS true. No exceptions.

The bigger the outlier, the bigger the gap between them.

This gap itself tells you how skewed your data is!

1 Arithmetic Mean — The Regular Average

What is it?

Add up ALL the values, then divide by the total count.

This is what most people mean when they say 'the average'.

Formula:

$$\text{Mean} = \frac{\text{Sum of all values}}{\text{Number of values}} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$$

Dataset: [10, 20, 20, 30, 40, 100]
Mean = (10 + 20 + 20 + 30 + 40 + 100) / 6
= 220 / 6
= 36.67

Real-Life Analogy

Six friends go out to eat. They pay: Rs. 10, Rs. 20, Rs. 20, Rs. 30, Rs. 40, Rs. 100.

Arithmetic Mean answers: 'If everyone had paid EQUALLY, how much would each person pay?'

Answer: Rs. 220 ÷ 6 = Rs. 36.67 each.

When to Use Mean

 USE Mean when...	 AVOID Mean when...
Data is symmetric (balanced left-right)	Outliers are present (they pull it badly)
No extreme outliers exist	Data is skewed
Class average marks	Income data (billionaires distort it)
Average temperature over a week	House prices in mixed neighbourhoods

The Outlier Problem — Why Mean Can Lie

Dataset: [10, 20, 20, 30, 40, 100]

Most students scored between 10 and 40. That is where the real story is.

But ONE student scored 100 — and the mean jumps to 36.67.

This is MISLEADING because 36.67 does not represent any student well.

Most students are way below it. One student is way above it.

This is why: when outliers exist, use Median instead.

Python Code

```
import pandas as pd
from scipy import stats as st

data = {"marks": [10, 20, 20, 30, 40, 100]}
df = pd.DataFrame(data)

arithmetic_mean = df["marks"].mean()    # 36.67
print(f"Arithmetic Mean: {arithmetic_mean:.2f}")

# Manual check:
print(f"Sum    = {df['marks'].sum()}")    # 220
print(f"Count  = {len(df)}")             # 6
print(f"Mean   = {df['marks'].sum()} / {len(df)} = {df['marks'].mean():.2f}")
```

2 Geometric Mean — For Growth & Ratios

What is it?

Multiply ALL values together, then take the nth root (where n = number of values).

This is the right tool when values MULTIPLY each other, not add.

Formula:

$$\text{Geometric Mean} = \sqrt[n]{x_1 \times x_2 \times x_3 \times \dots \times x_n}$$

Dataset: [10, 20, 20, 30, 40, 100]

$$\begin{aligned} \text{G. Mean} &= \sqrt[6]{10 \times 20 \times 20 \times 30 \times 40 \times 100} \\ &= \sqrt[6]{48,000,000,000} \\ &\approx 26.05 \end{aligned}$$

Real-Life Analogy — Investment Growth

Why Arithmetic Mean Gives Wrong Answer for Investments

Your money grows each year by different rates:

Year 1: +10% → multiply by 1.10

Year 2: +20% → multiply by 1.20

Year 3: +50% → multiply by 1.50

Wrong (Arithmetic Mean): $(10 + 20 + 50) / 3 = 26.67\%$ per year ← INCORRECT

Correct (Geometric Mean): $\sqrt[3]{1.10 \times 1.20 \times 1.50} - 1 \approx 25.99\%$ per year ← RIGHT

Why? Because your money COMPOUNDS (multiplies each year).

Arithmetic mean pretends the rates ADD — which is wrong for growth.

When to Use Geometric Mean

 USE Geometric Mean when...	 AVOID Geometric Mean when...
Investment returns / compound interest	Any value is zero (result = 0 always)
Population growth rates	Any value is negative
Averaging percentages and growth ratios	Values do not multiply together
Any time values MULTIPLY not ADD	General everyday averaging

Python Code

```
from scipy import stats as st

marks = [10, 20, 20, 30, 40, 100]

geometric_mean = st.gmean(marks)    # 26.05
print(f"Geometric Mean: {geometric_mean:.2f}")

# Compare with arithmetic mean:
print(f"Arithmetic Mean = {sum(marks)/len(marks):.2f}    ← higher (pulled by 100)")
print(f"Geometric Mean   = {geometric_mean:.2f}    ← lower, less affected by 100")

# Key rule: Geometric Mean is ALWAYS  $\leq$  Arithmetic Mean
```

3 Harmonic Mean — For Speeds & Rates

What is it?

Take the reciprocal ($1/x$) of each value, find their arithmetic mean, then take the reciprocal of that result.
In other words: n divided by the sum of ($1/\text{each value}$).

Formula:

$$\text{Harmonic Mean} = \frac{n}{1/x_1 + 1/x_2 + \dots + 1/x_n}$$

Dataset: [10, 20, 20, 30, 40, 100]

$$\begin{aligned} \text{H. Mean} &= 6 / (1/10 + 1/20 + 1/20 + 1/30 + 1/40 + 1/100) \\ &= 6 / (0.1 + 0.05 + 0.05 + 0.033 + 0.025 + 0.01) \\ &= 6 / 0.268 \\ &\approx 22.39 \end{aligned}$$

The Classic Analogy — Average Speed

The Speed Trap — Why Arithmetic Mean Fails Here

You drive from City A to City B at 60 km/h.

You return from City B to City A at 40 km/h.

Question: What is your AVERAGE speed for the whole trip?

✗ Wrong (Arithmetic): $(60 + 40) / 2 = 50$ km/h

✓ Correct (Harmonic): $2 / (1/60 + 1/40) = 48$ km/h

WHY is arithmetic wrong here?

You drive MORE TIME at the slower speed (40 km/h).

So 40 km/h has MORE INFLUENCE on the average.

Harmonic mean automatically accounts for this — arithmetic mean does not.

Rule of thumb: if distances are equal but speeds differ → use Harmonic Mean.

Real ML Use-Case — F1 Score

F1 Score in Machine Learning

F1 Score = Harmonic Mean of Precision and Recall

Precision = out of all predictions the model called positive, how many were actually positive?

Recall = out of all actual positives, how many did the model find?

$$F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

= Harmonic Mean of Precision and Recall

Why Harmonic? Because if one of them is 0, $F1 = 0$ — punishing a useless model.
Arithmetic mean would give a misleadingly high score even with one being 0.

✓ When to Use Harmonic Mean

✓ USE Harmonic Mean when...	✗ AVOID when...
Average speed over equal distances	Any value is zero
Price-to-earnings ratios in finance	Values are not rates or ratios
F1 Score in ML (precision & recall)	General everyday averaging
Averaging rates where denominator varies	You want to give weight to large values

Python Code

```
from scipy import stats as st

marks = [10, 20, 20, 30, 40, 100]

harmonic_mean = st.hmean(marks)    # 22.39
print(f"Harmonic Mean: {harmonic_mean:.2f}")

# The Three Means compared (rule:  $H \leq G \leq A$  always!):
print(f"Arithmetic Mean = {sum(marks)/len(marks):.2f}    ← largest")
print(f"Geometric Mean  = {st.gmean(marks):.2f}         ← middle")
print(f"Harmonic Mean   = {harmonic_mean:.2f}          ← smallest")

# F1 Score example:
precision = 0.8
recall    = 0.6
f1 = st.hmean([precision, recall])    # 0.6857
print(f"F1 Score: {f1:.4f}")
```

4 Median — The Robust Middle Value

What is it?

Sort all values in order, then pick the MIDDLE one.

If the count is even, average the two middle values.

```
Rule:
  Sort data → find the middle
  If n is ODD → middle value = position (n+1)/2
  If n is EVEN → average of positions n/2 and n/2+1

Dataset: [10, 20, 20, 30, 40, 100] → n = 6 (even)
Sorted:  10  20  20  30  40  100
           ↑    ↑
        Middle two: 20 and 30
Median = (20 + 30) / 2 = 25
```

Real-Life Analogy — House Prices

Why Newspapers Say 'Median Income' not 'Average Income'

Imagine a neighbourhood with 6 houses priced:

Rs. 10L, Rs. 20L, Rs. 20L, Rs. 30L, Rs. 40L, Rs. 5 Crore

Arithmetic Mean: $(10+20+20+30+40+500) / 6 = 103$ Lakh

Median: $(20 + 30) / 2 = 25$ Lakh

Which represents a TYPICAL house better? Rs. 103L or Rs. 25L?

Obviously Rs. 25L — because most houses are in that range.

The one Rs. 5 Crore mansion completely breaks the mean.

This is exactly why income statistics use MEDIAN.

A few billionaires would make the 'average' citizen look rich — which is false.

When to Use Median

 USE Median when...	 AVOID Median when...
Data has outliers	You need to use all values in further math
Data is skewed (right or left)	Data is symmetric with no outliers (use mean instead)
Income and salary reports	You need sensitivity to all values
House prices	

Key Insight — Median vs Mean Reveals Skewness

This is a powerful and practical trick:

- Mean $\approx$ Median → data is likely SYMMETRIC (balanced)
- Mean > Median → data is likely RIGHT-SKEWED (big values pulling mean right)
- Mean < Median → data is likely LEFT-SKEWED (small values pulling mean left)

Your dataset [10, 20, 20, 30, 40, 100]:

Mean = 36.67

Median = 25

Mean > Median → RIGHT-SKEWED ✓ (100 pulls the mean to the right)

Python Code

```
import pandas as pd

marks = [10, 20, 20, 30, 40, 100]
df = pd.DataFrame({"marks": marks})

median = df["marks"].median()    # 25.0
print(f"Median: {median}")

# See how median ignores the outlier:
print(f"Mean    = {df['marks'].mean():.2f}    ← pulled up to 36.67 by 100")
print(f"Median = {median}                    ← not affected by 100 at all")

# Check skewness: mean > median → right-skewed
if df["marks"].mean() > median:
    print("Data is RIGHT-SKEWED (tail on the right)")
elif df["marks"].mean() < median:
    print("Data is LEFT-SKEWED (tail on the left)")
else:
    print("Data is approximately SYMMETRIC")
```

5 Mode — The Most Popular Value

What is it?

The value that appears MOST FREQUENTLY in the dataset.

It is the only measure of central tendency that can be used for non-numeric (categorical) data.

```
Dataset: [10, 20, 20, 30, 40, 100]

Value  Count
  10     1
  20     2  ← appears most → Mode = 20
  30     1
  40     1
 100     1

Mode = 20
```

Real-Life Analogy — Shoe Store

Shoe Store Example

A shoe store wants to decide which size to ORDER THE MOST.

They cannot use arithmetic mean — size 7.3 does not exist!

They cannot use median — they need to know what people actually buy most.

They need the MODE — the most frequently sold size.

If size 9 sells 500 times and all other sizes sell 50 times each, they should stock mostly size 9. That is the mode.

Mode is used when you need the most POPULAR or most COMMON value.

Special Cases — No Mode, Multiple Modes

Case	Example	What it means
No mode	[1, 2, 3, 4, 5]	All values unique — no value repeats
Unimodal	[1, 2, 2, 3, 4]	One mode (2) — most common
Bimodal	[1, 2, 2, 3, 3]	Two modes (2 and 3) — two popular values
Categorical	[Red, Blue, Blue, Green]	Mode = Blue — works on non-numbers too!

✓ When to Use Mode

✓ USE Mode when...	✗ AVOID Mode when...
Finding most popular product in a store	All values are unique (no mode exists)
Most common shirt size to manufacture	You need to account for magnitude
Survey about favourite colour, category	Continuous numeric data with no repeats
ONLY measure that works on category data	

💻 Python Code

```
import pandas as pd
from scipy import stats as st

marks = [10, 20, 20, 30, 40, 100]
df = pd.DataFrame({"marks": marks})

# Method 1: Pandas
mode_pandas = df["marks"].mode()[0]          # 20
print(f"Mode (pandas): {mode_pandas}")

# Method 2: scipy
mode_scipy = st.mode(marks).mode             # 20
print(f"Mode (scipy) : {mode_scipy}")

# See value counts:
print("\nValue counts:")
print(df["marks"].value_counts().sort_index())

# Categorical mode example:
colors = pd.Series(["Red", "Blue", "Blue", "Green", "Blue"])
print(f"\nFavourite colour: {colors.mode()[0]}")  # Blue
```

6 Skewness — Symmetric, Right-Skewed & Left-Skewed

💡 What is Skewness?

Skewness tells us whether the data is **BALANCED** on both sides or whether it **LEANS** to one side. It describes the **SHAPE** of your data distribution.

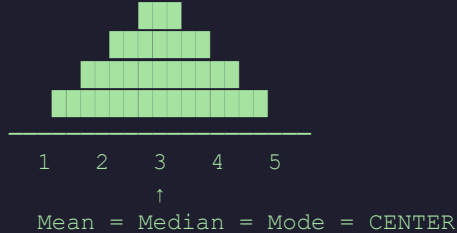
The **KEY** concept: the **TAIL**.

A tail is the long, stretched, thin part of the graph where a **FEW** extreme values reach far away. You name the skew after **WHERE** the tail points.

🟦 Symmetric Data — Perfectly Balanced

Example: `[1, 2, 3, 4, 5]` or any bell-shaped data

Shape:



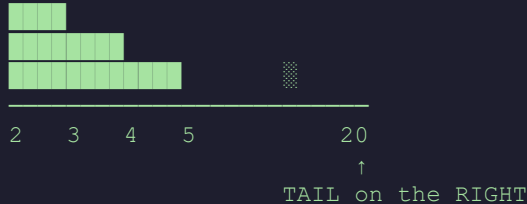
Left side mirrors the right side exactly.
Mean $\approx$ Median ← quick check for symmetry

➡ Right-Skewed (Positive Skew) — Tail on the RIGHT

Most values are **SMALL**. A few very **LARGE** values stretch to the right, forming the tail.

Example: `[2, 3, 4, 5, 20]` or income data

Shape:



Key rules:

Mean > Median (mean gets pulled toward the big outlier)
Tail goes RIGHT = Right-Skewed

Real-life examples:

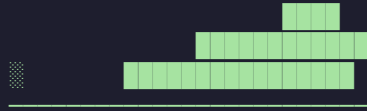
Income data — most earn normal, a few billionaires earn crores
House prices — most moderate, a few mansions are extreme
Your dataset `[5, 7, 8, 9, 10, 100]` → RIGHT-SKEWED (mean=23, median=8.5)

← Left-Skewed (Negative Skew) — Tail on the LEFT

Most values are LARGE. A few very SMALL values stretch to the left, forming the tail.

Example: [1, 10, 11, 12, 13] or easy exam scores

Shape:



1 10 11 12 13

↑

TAIL on the LEFT

Key rules:

Mean < Median (mean gets pulled toward the small outlier)

Tail goes LEFT = Left-Skewed

Real-life examples:

Easy exam — most students score high, a few fail badly

Age at retirement — most retire around 60, a few retire very early

Distribution	Tail Direction	Mean vs Median	Real-World Example
Symmetric	None (balanced)	Mean $\approx$ Median	Heights, IQ scores, bell curve
Right-Skewed	→ Right	Mean > Median	Income, house prices, your [5,7,8,9,10,100]
Left-Skewed	← Left	Mean < Median	Easy exam scores, age at retirement

🧠 The Easy Trick to Remember Skewness Forever

Forget which side is the 'bulk'. Just look where the WEIRD values go.

Imagine data as a crowd of people on a number line.

The tail = one person standing FAR away from everyone else.

Which SIDE is that person on?

Right side → Right-Skewed

Left side → Left-Skewed

And remember: Mean gets PULLED toward that person (the outlier).

So the mean chases the tail.

Are Skewness and Outliers the Same?

Related — but NOT identical. This is a common beginner confusion.

Skewness	Outliers
SHAPE of the overall distribution	Individual EXTREME VALUES far from the rest
Measured by comparing mean vs median	Detected using IQR bounds
Can exist without extreme individual points	Can exist even if data is roughly symmetric

Skewed data OFTEN has outliers — but you can have skew without extreme individual points.
And you can have outliers on BOTH ends that balance out, making distribution look symmetric.

7 All 5 Measures — Side by Side Comparison

 Dataset: [10, 20, 20, 30, 40, 100]

Measure	Value	Best For	Handles Outlier (100)?
Arithmetic Mean	36.67	General average, symmetric data	✗ No — pulled up to 36.67 by the outlier
Geometric Mean	26.05	Growth rates, ratios	⚠ Somewhat — still affected but less
Harmonic Mean	22.39	Speeds, rates, F1 score	✓ Better — gives lowest weight to large values
Median	25.00	Skewed data, outlier-heavy data	✓ Yes — median stays at 25 regardless of outlier
Mode	20	Most frequent value, categories	✓ Yes — not affected at all

💡 Key Rule (Always True):

$$\begin{array}{ccccc} \text{Harmonic Mean} & \leq & \text{Geometric Mean} & \leq & \text{Arithmetic Mean} \\ 22.39 & \leq & 26.05 & \leq & 36.67 \end{array}$$

The bigger the outlier, the BIGGER the gap between these three.
The gap itself is a signal of how skewed your data is!

Complete Python Code — All 5 Measures

```
import pandas as pd
from scipy import stats as st

data = {"marks": [10, 20, 20, 30, 40, 100]}
df = pd.DataFrame(data)
marks = df["marks"]

arith = marks.mean()           # 36.67
geo = st.gmean(marks)         # 26.05
harm = st.hmean(marks)        # 22.39
med = marks.median()          # 25.00
mode = marks.mode()[0]        # 20

summary = pd.DataFrame({
    "Measure" : ["Arithmetic Mean", "Geometric Mean", "Harmonic Mean", "Median", "Mode"],
    "Value"   : [round(arith,2), round(geo,2), round(harm,2), med, mode],
    "Best For": [
        "General average, symmetric data",
```

```
        "Growth rates, ratios, percentages",
        "Speeds, rates, ratios",
        "Skewed data, data with outliers",
        "Most frequent value, categories"
    ],
    "Outlier Safe": ["No", "Somewhat", "Better", "Yes", "Yes"]
})

print(summary.to_string(index=False))
print()
print("Rule: Harmonic  $\leq$  Geometric  $\leq$  Arithmetic (always true)")
print(f"        {harm:.2f}  $\leq$  {geo:.2f}  $\leq$  {arith:.2f}")
```


8 Decision Guide — When to Use Which Measure

Quick Decision Tree:

Is your data CATEGORICAL (colours, names, categories)?
→ MODE (only choice – you cannot average categories)

Is your data numeric?
Does it involve RATES or SPEEDS (time per unit, km/h)?
→ HARMONIC MEAN

Does it involve GROWTH or MULTIPLICATION (interest, population %)?
→ GEOMETRIC MEAN

Does it have OUTLIERS or is it SKEWED?
→ MEDIAN (robust, not pulled by extremes)

Is it symmetric, balanced, no extreme outliers?
→ ARITHMETIC MEAN (standard, intuitive)

Measure	Scenario	Why
Arithmetic Mean	Class test average (no outliers)	All marks are in similar range, balanced data
Geometric Mean	Average investment return over 5 years	Returns compound (multiply) each year
Harmonic Mean	Average speed of a round trip	Speed and time have inverse relationship
Median	Typical house price in a city	A few luxury homes would destroy the mean
Mode	Most popular shoe size to stock	Need the MOST COMMON value, not average

Final Cheat Sheet — Everything in One Page

Concept	One-Line Definition	Real-World Example
Arithmetic Mean	Sum of all values $\div$ count	Average marks in a class
Geometric Mean	n th root of product of all values	Average investment return rate
Harmonic Mean	$n \div (\text{sum of reciprocals})$	Average speed on a round trip
Median	Middle value after sorting	Typical house price in a city
Mode	Most frequently occurring value	Most popular shoe size to stock
Symmetric data	Both sides mirror each other	Heights of people, IQ scores
Right-skewed	Tail on right, mean $>$ median	Income data, house prices
Left-skewed	Tail on left, mean $<$ median	Easy exam scores
Outlier	Value far outside IQR bounds	One billionaire in a salary dataset
$H \leq G \leq A$ rule	Harmonic $\leq$ Geometric $\leq$ Arithmetic — ALWAYS	$22.39 \leq 26.05 \leq 36.67$ for your dataset



Your Learning Journey So Far

You have now covered:

- ✓ Central Tendency — how to find the 'center' of data (this notebook)
- ✓ Arithmetic Mean, Geometric Mean, Harmonic Mean, Median, Mode
- ✓ Skewness — symmetric, right-skewed, left-skewed
- ✓ Why mean fails with outliers and when to use median instead
- ✓ The $H \leq G \leq A$ rule

Next up: Measurement of Dispersion $\rightarrow$ how SPREAD OUT your data is

- $\rightarrow$ Range, IQR, Variance, Standard Deviation, Standard Error
- $\rightarrow$ Outlier detection and data cleaning
- $\rightarrow$ Hypothesis Testing and Confidence Intervals

You are building a SOLID foundation for Data Science and ML. Keep going!